

Semester Project

Project Title

Organization-Based Web Application using Python Flask

Project Objective

The objective of this project is to enable students to:

- Apply **HTML, CSS, JavaScript, Python & Flask**
- **Database usage IS mandatory** (any type of database work)
- Build a **real-world organization system**
- Integrate a **basic chatbot**
- Create a **personal student portfolio page**

Project Description

Each student must develop a **complete web application for an organization** (real or imaginary).

The system must store data in a **database**, include a **chatbot**, and contain a **student portfolio page**.

Note: This project must be developed individually and submitted in the upcoming theory class.

Mandatory Project Modules

1. Organization Website Module

The project must represent an organization such as:

- School / College
- Training Institute
- Software House
- NGO
- Hospital
- Company Management System

Required Pages:

- Home
- About Organization
- Services / Departments
- Contact Page

2. Database Module (Mandatory)

- Use **SQLite database**
- Database must store **real and meaningful data**
- **Full CRUD operations are NOT mandatory**

✓ Acceptable Database Usage Examples:

Students may use the database for **any one or more** of the following purposes:

- Storing contact form messages
- Saving chatbot user queries and responses
- Storing login / registration data
- Saving student or employee profiles
- Storing appointments or registrations
- Storing feedback or reviews
- Saving project activity logs

Note: At least **one database interaction (INSERT / SELECT)** must be implemented successfully.

3. Chatbot Module (Mandatory)

The project must include a **basic chatbot** that can:

- Answer FAQs about the organization
- Respond to predefined questions
- Be rule-based or simple AI logic

Examples:

- “What courses are offered?”
- “Office timing?”
- “Contact details?”

(Chatbot can be simple — intelligence is not graded, **implementation is.**)

4. Student Portfolio Page (Mandatory)

Each project must include a **personal portfolio page** of the student.

Portfolio Must Include:

- Student Name
- Roll Number
- Department / Semester

- Skills
- Projects
- Contact Information
- Profile Picture (optional)

5. Authentication (Recommended)

- Login system (Admin / User)
- Session handling (optional but encouraged)

Technology Stack (Mandatory)

Layer	Technology
Frontend	HTML, CSS, JavaScript
Backend	Python Flask
Database	SQLite

Folder Structure (Recommended)

```

project/
├── app.py
├── database.db
├── static/
│   └── style.css
├── templates/
│   ├── index.html
│   ├── portfolio.html
│   ├── chatbot.html
│   └── dashboard.html
└── README.txt

```

Documentation Requirements

Each student must submit a **printed project report** containing:

1. Project Title
2. Student Information
3. Problem Statement
4. Objectives
5. Tools & Technologies
6. System Features
7. Database Design
8. Screenshots
9. Conclusion

Rules & Regulations

- ✗ Group work is **NOT allowed**
- ✗ Copy-paste projects will receive **zero marks**
- ✗ Late submission will not be accepted
- ✓ Original work is mandatory
- ✓ Viva questions may be asked

Submission Instructions

- Project must be submitted in **upcoming theory class**
- Bring:
 - Printed documentation
 - Laptop with working project
- Project must **run successfully on local machine**